

Лабораторная работа

Номер 6

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Информация

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Развить навыки администрирования ОС Linux. Получить первое практическое знакомство с технологией SELinux. Проверить работу SELinux на практике совместно с веб-сервером Apache.

Проверим, что SELinux работает в режиме enforcing

```
[nsandriushin@nsandriushin ~]$ getenforce
Enforcing
[nsandriushin@nsandriushin ~]$ sestatus
SELinux status:                enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:              targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
[nsandriushin@nsandriushin ~]$ service httpd status
Redirecting to /bin/systemctl status httpd.service
○ httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:httpd.service(8)
[nsandriushin@nsandriushin ~]$ service httpd start
Redirecting to /bin/systemctl start httpd.service
[nsandriushin@nsandriushin ~]$ ps aux | grep httpd
system_u:system_r:httpd_t:s0  root      42803  0.3  0.2 21032 11372 ?        Ss   21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0  apache  42813  0.0  0.1 22908  7384 ?        S    21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0  apache  42815  1.0  0.4 2227512 17172 ?    Sl   21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0  apache  42816  0.6  0.3 2358648 15164 ?    Sl   21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0  apache  42817  0.8  0.4 2227512 17180 ?    Sl   21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
unconfined_u:unconfined_r:unconfined_t:s0-c0.c1023 nsandri+ 43017 0.0  0.0 221664 2432 pts/0 S+  21:38
0:00 grep --color=auto httpd
```

Рис. 1: Режим SELinux

Посмотрим переключатели SELinux для httpd

```
[nsandrushin@nsandrushin ~]$ sestatus -b | grep httpd
httpd_anon_write          off
httpd_builtin_scripthing  on
httpd_can_check_spam      off
httpd_can_connect_ftp     off
httpd_can_connect_ldap    off
httpd_can_connect_mythtv  off
httpd_can_connect_rabbitmq off
httpd_can_manage_courier_spool off
httpd_can_network_connect off
httpd_can_network_connect_cobbler off
httpd_can_network_connect_db off
httpd_can_network_memcache off
httpd_can_network_relay   off
httpd_can_sendmail        off
httpd_dbus_avahi          off
httpd_dbus_ssd            off
httpd_dontaudit_search_dirs off
httpd_enable_cgi          on
httpd_enable_ftp_server   off
httpd_enable_homedirs     off
httpd_execmem             off
httpd_graceful_shutdown  off
httpd_manage_ipa          off
httpd_mod_auth_ntlm_winbind off
httpd_mod_auth_pam        off
httpd_read_user_content   off
httpd_run_ipa             off
httpd_run_preupgrade      off
httpd_run_stickshift      off
httpd_serve_cobbler_files off
httpd_setrlimit           off
httpd_ssi_exec            off
httpd_sys_script_anon_write off
httpd_ttp_exec            off
httpd_tty_comm            off
httpd_unified             off
httpd_use_cifs            off
httpd_use_fusefs          off
httpd_use_gpg             off
httpd_use_ifs             off
httpd_use_openssl          off
httpd_use_openssl         off
httpd_use_openssl         off
httpd_use_openssl         off
httpd_use_openssl         off
httpd_use_openssl         off
httpd_verify_dns          off
```

Рис. 2: Переключатели SELinux

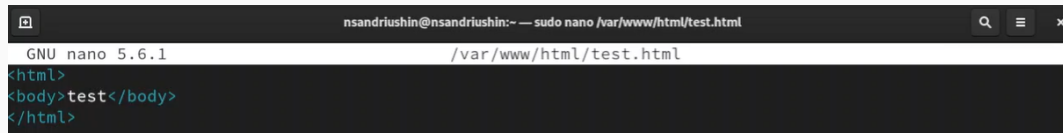
Посмотрим на количество типов, пользователей и ролей с помощью команды seinfo

```
[nsandriushin@nsandriushin ~]$ seinfo
Statistics for policy file: /sys/fs/selinux/policy
Policy Version:          33 (MLS enabled)
Target Policy:           selinux
Handle unknown classes:  allow
Classes:                 135
Sensitivities:           1
Types:                   5169
Users:                   8
Booleans:                358
Allow:                   65633
Auditallow:              176
Type_trans:              271851
Type_member:             37
Role_allow:              40
Constraints:             70
MLS Constrain:           72
Permissives:             1
Defaults:                7
Allowxperm:              0
Auditallowxperm:         0
Ibendportcon:            0
Initial SIDs:            27
Genfscon:                109
Netifcon:                0
Permissions:             457
Categories:             1024
Attributes:              259
Roles:                   15
Cond. Expr.:            390
Neverallow:              0
Dontaudit:               8703
Type_change:             94
Range_trans:             5931
Role_trans:              417
Validatetrans:           0
MLS Val. Tran:           0
Polcap:                  6
Typebounds:              0
Neverallowxperm:         0
Dontauditxperm:          0
Ibpkeycon:               0
Fs_use:                  35
Portcon:                 665
Nodecon:                 0

[nsandriushin@nsandriushin ~]$ ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Jan 22 03:25 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0 6 Jan 22 03:25 html
[nsandriushin@nsandriushin ~]$ ls -lZ /var/www/html
total 0
[nsandriushin@nsandriushin ~]$ sudo nano /var/www/html/test.html
```

Рис. 3: seinfo

Создадим файл /var/www/html/test.html и заполним его одним словом



```
nsandriushin@nsandriushin:~ — sudo nano /var/www/html/test.html
GNU nano 5.6.1 /var/www/html/test.html
<html>
<body>test</body>
</html>
```

Рис. 4: /var/www/html/test.html

При открытии браузера открывается созданная нами страница

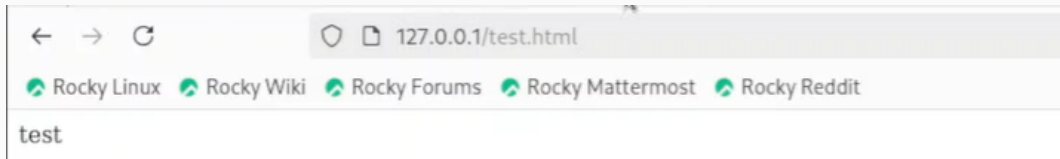


Рис. 5: Открытие страницы

После смены метки доступ к странице был запрещён

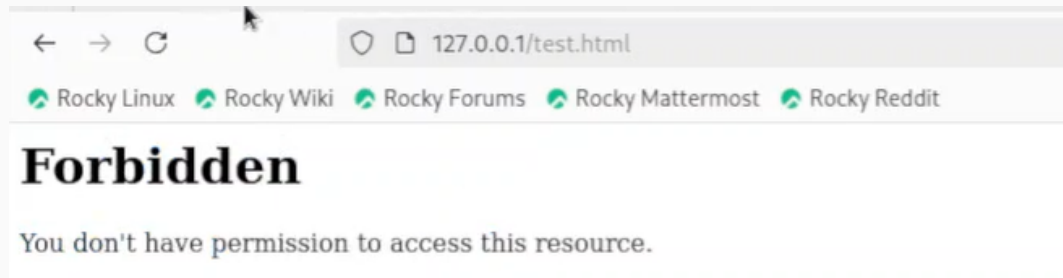


Рис. 7: Доступ запрещён

Поменяем порт на 81 вместо 80

```
GNU nano 5.6.1 /etc/httpd/conf/httpd.conf Modified
#
# This is the main Apache HTTP server configuration file. It contains the
# configuration directives that give the server its instructions.
# See http://httpd.apache.org/docs/2.4/ for detailed information.
# In particular, see
# <URL>http://httpd.apache.org/docs/2.4/mod/directives.html
# for a discussion of each configuration directive.
#
# See the httpd.conf(5) man page for more information on this configuration,
# and httpd.service(8) on using and configuring the httpd service.
#
# Do NOT simply read the instructions in here without understanding
# what they do. They're here only as hints or reminders. If you are unsure
# consult the online docs. You have been warned.
#
# Configuration and logfile names: If the filenames you specify for many
# of the server's control files begin with '/' (or 'drive:/' for win32), the
# server will use that explicit path. If the filenames do 'not' begin
# with '/', the value of ServerRoot is prepended -- so 'log/access_log'
# with ServerRoot set to '/www' will be interpreted by the
# server as '/www/log/access_log', where as '/log/access_log' will be
# interpreted as '/log/access_log'.
#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path. If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used. If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts. See the httpd.service(8) man
# page for more information.
#
Listen 12.34.56.78:80
Listen 81
```

Рис. 8: Смена порта

Перезагрузим службу. После этого мы не сможем зайти на нашу страницу. Об этом также будут писать логи

```
(nsandrushin@nsandrushin ~)$ sudo curl -Jc https://api.github.com/
(nsandrushin@nsandrushin ~)$ systemctl restart httpd
(nsandrushin@nsandrushin ~)$ service httpd restart
Redirecting to /bin/systemctl restart httpd.service
(nsandrushin@nsandrushin ~)$ tail -n1 /var/log/messages
tail: invalid number of lines: '1'
(nsandrushin@nsandrushin ~)$ sudo tail -n1 /var/log/messages
tail: invalid number of lines: '1'
(nsandrushin@nsandrushin ~)$ sudo tail -n /var/log/messages
tail: invalid number of lines: '/var/log/messages'
(nsandrushin@nsandrushin ~)$ sudo curl /var/log/messages
May  3 21:53:10 nsandrushin systemd[1]: Started the Apache HTTP server.
May  3 21:53:12 nsandrushin systemd[1]: Starting dnf makecache...
May  3 21:53:13 nsandrushin dnf[44307]: Rocky Linux 9 - BaseOS
00:00
May  3 21:53:13 nsandrushin dnf[44307]: Rocky Linux 9 - AppStream
00:00
May  3 21:53:14 nsandrushin dnf[44307]: Rocky Linux 9 - Extras
0.5 kB/s | 2.9 MB
May  3 21:53:14 nsandrushin dnf[44307]: Metadata cache created.
May  3 21:53:14 nsandrushin systemd[1]: dnf-makecache.service: Deactivated successfully.
May  3 21:53:14 nsandrushin systemd[1]: Finished dnf makecache.
May  3 21:53:14 nsandrushin systemd[1]: dnf-makecache.service: Consumed 1.182s CPU time.
May  3 21:53:23 nsandrushin gnome-shell[1844]: libinput error: client bug: timer event4 debounce short: a
libinput event4 is in the past (-20ms), your system is too slow
(nsandrushin@nsandrushin ~)$ sudo tail /var/log/httpd/error_log
tail: cannot open '/var/log/httpd/error_log' for reading: No such file or directory
(nsandrushin@nsandrushin ~)$ sudo tail /var/log/auth.log
type=USER_AUTH msg=audit(1174628481.844:356): pid=4444 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0-c1023 msg=op=PAK:accounting granter=psm_unix pam_localuser acct=nsandrushi
n exe="/usr/bin/sudo" hostname=1 addr=1 terminal=/dev/pts/0 res=success UID="nsandrushin" AUID="nsandrui
shin"
type=USER_CMD msg=audit(1174628481.844:356): pid=4444 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0-c1023 msg=cwd="/home/nsandrushin" cmd=7461596C2027F61727F0C8F672F687474702F
672720F725F0C8F672F687474702F672720F725F0C8F672F687474702F672720F725F0C8F672F687474702F672720F725F0C8F672F687474702F
type=CRS_START msg=audit(1174628481.844:357): pid=4444 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0-c1023 msg=op=PAK:setcred granter=psm_unix pam_fprintd acct="root" exe="/usr/b
in/sudo" hostname=1 addr=1 terminal=/dev/pts/0 res=success UID="nsandrushin" AUID="nsandrushin"
type=USER_START msg=audit(1174628481.844:358): pid=4444 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0-c1023 msg=op=PAK:session_open granter=psm_unix pam_keyinit pam_libtasn1 pam_systemd pa
m_unix acct="root" exe="/usr/bin/sudo" hostname=1 addr=1 terminal=/dev/pts/0 res=success UID="nsandrushin"
AUID="nsandrushin"
type=USER_CMD msg=audit(1174628481.121:359): pid=4444 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0-c1023 msg=op=PAK:session_close granter=psm_unix pam_keyinit pam_libtasn1 pam_systemd pa
m_unix acct="root" exe="/usr/bin/sudo" hostname=1 addr=1 terminal=/dev/pts/0 res=success UID="nsandrushin"
AUID="nsandrushin"
type=CRS_STOP msg=audit(1174628481.126:360): pid=4444 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0-c1023 msg=op=PAK:setcred granter=psm_unix pam_fprintd acct="root" exe="/usr/b
in/sudo" hostname=1 addr=1 terminal=/dev/pts/0 res=success UID="nsandrushin" AUID="nsandrushin"
type=USER_AUTH msg=audit(1174628486.420:361): pid=4445 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0-c1023 msg=op=PAK:accounting granter=psm_unix pam_localuser acct=nsandrushi
n exe="/usr/bin/sudo" hostname=1 addr=1 terminal=/dev/pts/0 res=success UID="nsandrushin" AUID="nsandrui
shin"
type=USER_CMD msg=audit(1174628486.420:362): pid=4445 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
```

Рис. 9: Перезагрузка службы

Добавление порта и возврат к первоначальной конфигурации

Для того, чтобы получилось зайти, досаточно добавить 81 порт в semanage.
После Этого вернём всё как было

```
[nsandriushin@nsandriushin ~]$ sudo semanage port -a -t http_port_t -p tcp 81
Port tcp/81 already defined, modifying instead
[nsandriushin@nsandriushin ~]$ semanage port -l | grep http_port_t
ValueError: SELinux policy is not managed or store cannot be accessed.
[nsandriushin@nsandriushin ~]$ sudo semanage port -l | grep http_port_t
http_port_t                tcp      81, 80, 81, 443, 488, 8008, 8009, 8443, 9000
pegasus_http_port_t        tcp      5988
[nsandriushin@nsandriushin ~]$ service httpd restart
Redirecting to /bin/systemctl restart httpd.service
[nsandriushin@nsandriushin ~]$ chcon -t httpd_sys_content_t /var/www/html/test.html
chcon: failed to change context of '/var/www/html/test.html' to 'unconfined_u:object_r:httpd_sys_content_t:s0': Operation not permitted
[nsandriushin@nsandriushin ~]$ sudo chcon -t httpd_sys_content_t /var/www/html/test.html
[nsandriushin@nsandriushin ~]$ sudo nano /etc/httpd/conf/httpd.conf
[nsandriushin@nsandriushin ~]$ sudo semanage port -d -t http_port_t -p tcp 81
[nsandriushin@nsandriushin ~]$ sudo rm /var/www/html/test.html
[nsandriushin@nsandriushin ~]$
```

Рис. 10: Добавление порта и возврат к первоначальной конфигурации

При добавленном порте сайт работает, но нам нужно указать, что мы обращаемся именно к 81 порту

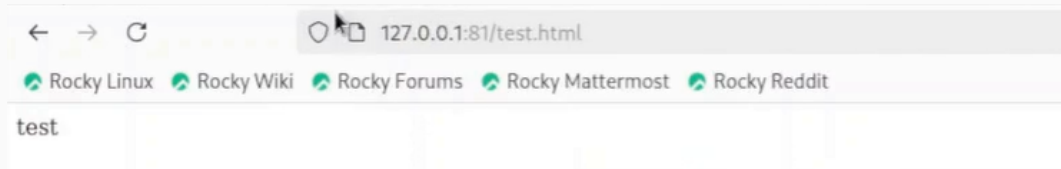


Рис. 11: Работающий сайт

Смена порта обратно

Меняем теперь порт отбратно

```
GNU nano 2.8.1 /etc/httpd/conf/httpd.conf Modified
#
# This is the main Apache HTTP server configuration file.  It contains the
# configuration directives that give the server its instructions.
# See http://httpd.apache.org/docs/2.4/ for detailed information.
# In particular, see
# http://httpd.apache.org/docs/2.4/mod/directives.html
# for a discussion of each configuration directive.
#
# See the httpd.conf(5) man page for more information on this configuration,
# and httpd.service(8) on using and configuring the httpd service.
#
# Do NOT simply read the instructions in here without understanding
# what they do.  They're here only as hints or reminders.  If you are unsure
# consult the online docs.  You have been warned.
#
# Configuration and logfile names: If the filenames you specify for many
# of the server's control files begin with "/" (or "drive/" for win32), the
# server will use that explicit path.  If the filenames do *not* begin
# with "/", the value of ServerRoot is prepended -- so 'log/access_log'
# with ServerRoot set to '/www' will be interpreted by the
# server as '/www/log/access_log', where as '/log/access_log' will be
# interpreted as '/log/access_log'.
#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path.  If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used.  If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default.  See also the VirtualHost
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts.  See the httpd.service(8) man
# page for more information.
#
Listen 12.34.56.78:80
Listen 80
```

Рис. 12: Смена порта обратно

В результате выполнения лабораторной работы было получено понимание работы мандатного разграничения доступа